Multi-Region AI-Powered Banking Application

Engineering Mission-Critical, High-Availability Cloud Infrastructure with Zero-Downtime Tolerance

Architecture: Active-Active DR

Tools: Terraform & Github Actions

Author: Ikenna Anasieze - Cloud & AI Engineer

Repository: Github.com/donaldnaz/Financeapp

1. Overview

Modern banking systems operate under rigorous availability frameworks. Even brief structural disruptions can inflict significant commercial damage, invite regulatory penalties, and compromise consumer confidence. This project documents the end-to-end design, implementation, and orchestration of a resilient, AI-powered banking application deployed on Amazon Web Services (AWS).

The system features a multi-region deployment architecture designed to guarantee immediate failover recovery without human intervention, ensuring continuous availability during widespread cloud outages. Every infrastructure asset is declared as code, allowing rapid deployment, predictable testing environments, and automated decommissioning.

2. Problem to Solve

Traditional financial monolithic systems and naive single-region cloud architectures are vulnerable to systemic regional downtime, or regional cloud service disruptions. The design criteria for this system were framed around eliminating two primary vulnerabilities:

- **Recovery Time Objective (RTO):** The maximum acceptable duration of downtime before service restoration. In mission-critical environments, the target metric must approach RTO → 0.
- **Recovery Point Objective (RPO):** The maximum acceptable age of data that can be lost due to an outage. For financial ledgers, transactional consistency requires an explicit target of RPO = 0.

3. Technology Stack

The stack incorporates specialized tools segmented into logical infrastructure layers to ensure structural rigidity and separation of concerns:

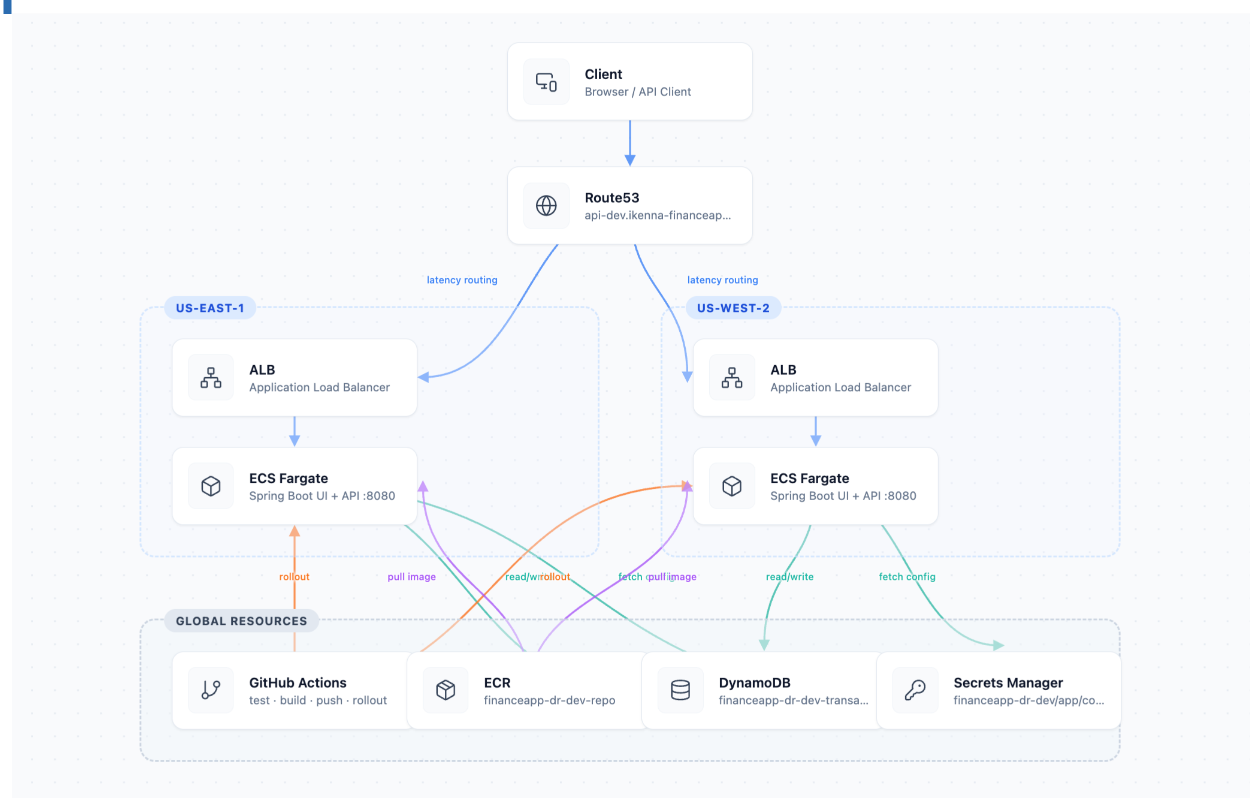
Infrastructure Layer	Technology Utilized	Functional Role
Cloud Provider	*Amazon Web Services (AWS)*	Global cloud infrastructure, cross-region networking, and managed abstractions.
Infrastructure as Code	*Terraform*	Declarative state files, modular network blocks, and multi-provider regional targeting.
CI/CD Engine	*GitHub Actions*	Automated build execution, application containerization, and structural testing

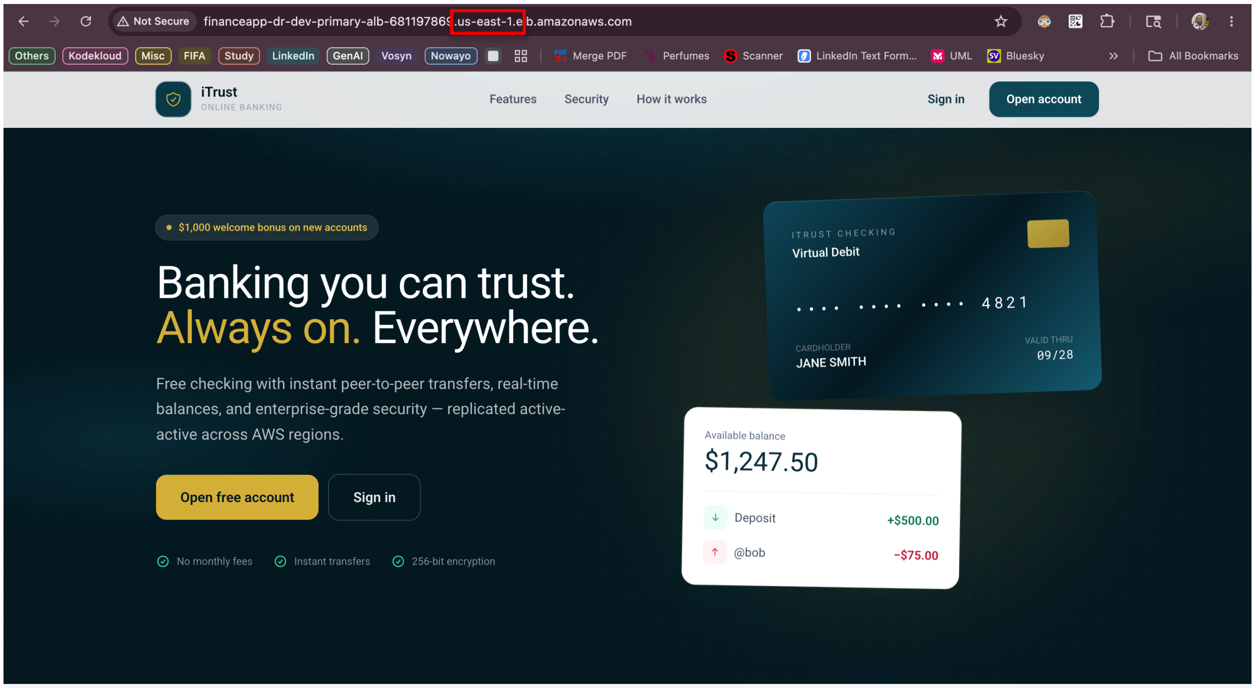
		pipelines.
Compute Tier	*AWS ECS / Fargate*	Serverless container execution, horizontal scaling, and isolated runtime layers.
Data Storage	*Amazon DynamoDB*	Global Tables implementation providing active-active cross-region multi-master synchronization.
Traffic Management	*AWS ALB & Route 53*	routing, continuous health profiling, and DNS failover record automation.
Artificial Intelligence	*Cursor IDE*	Natural Language processing for conversational banking, anomaly detection, and insights.

4. Architectural Flow

The architecture establishes isolation across two distinct geographic zones: us-east-1 (Primary Region) and us-west-2 (Secondary Region).

> Client Request → Route 53 (Health-Monitored) → Regional ALB → ECS Fargate Microservices → DynamoDB Global Tables ↔ Multi-Region Cross-Replication.

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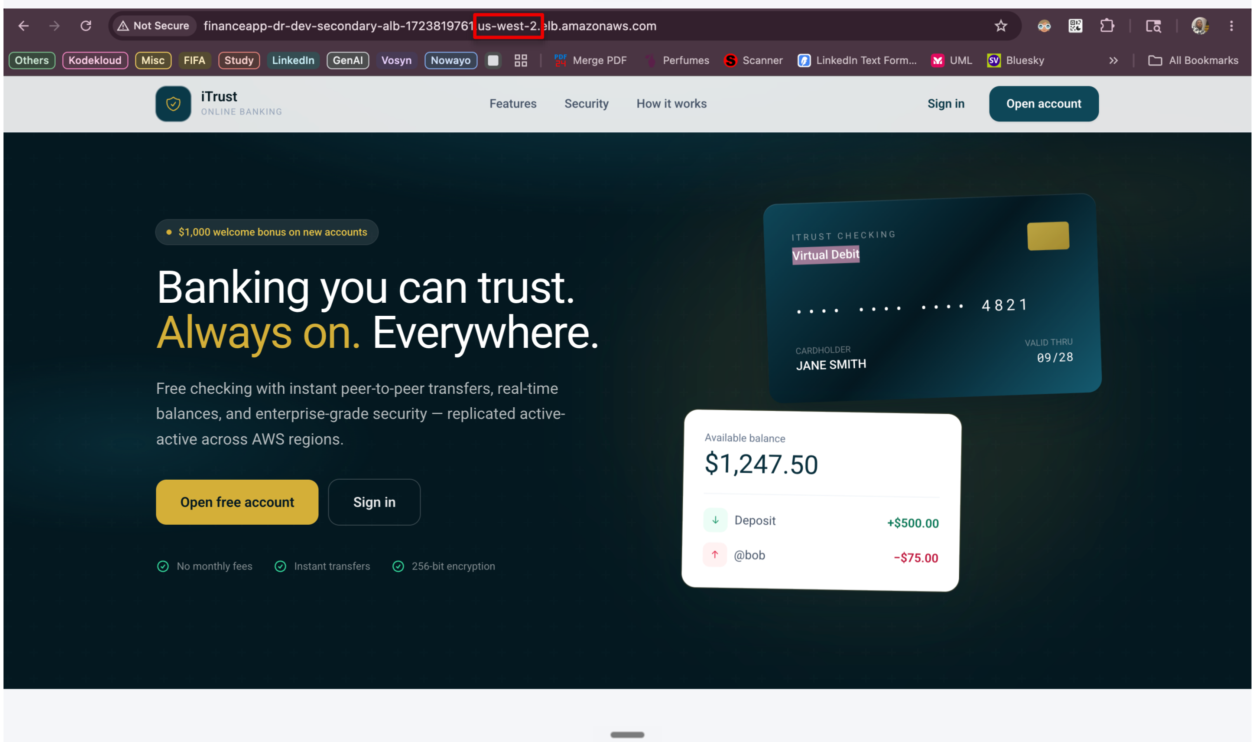
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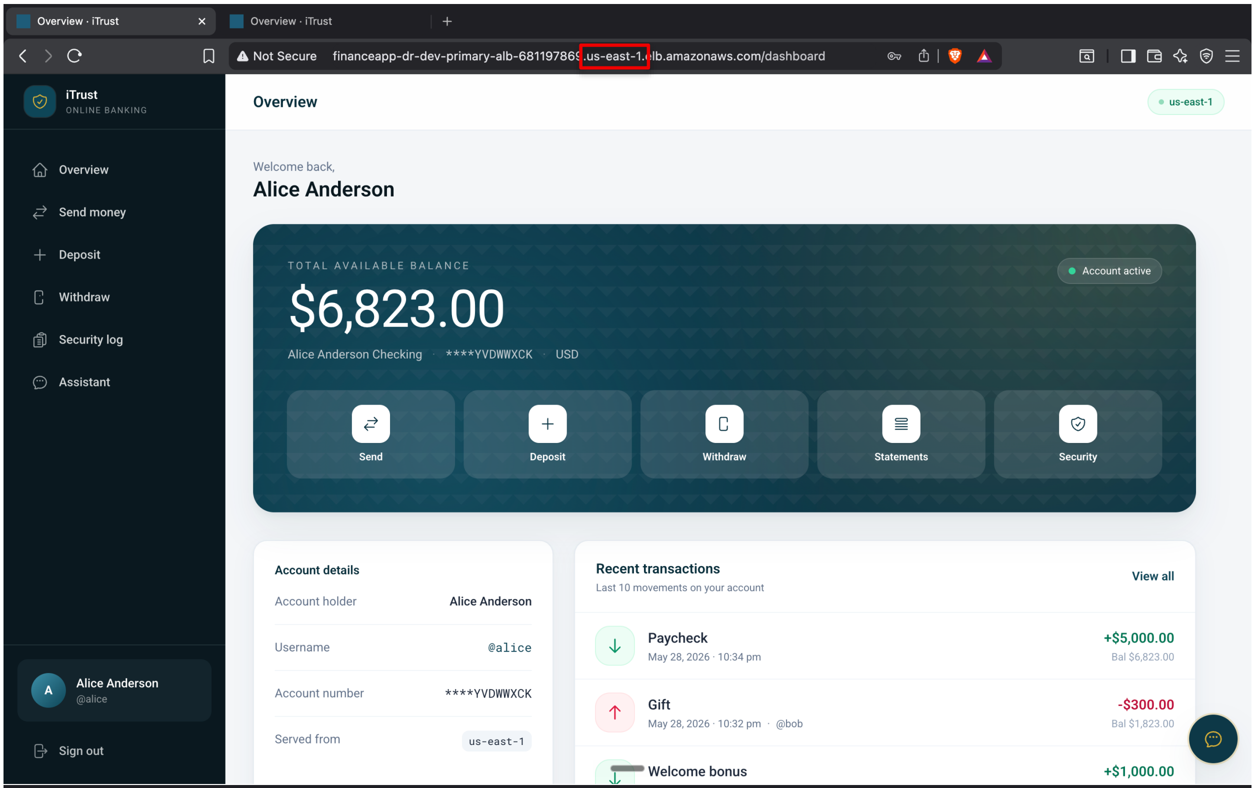
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Account holder Alice Anderson
Username @alice
Account number ****YVDWWXCK
Served from us-east-1
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View all
Paycheck
May 28, 2026 · 10:34 pm
+$5,000.00
Bal $6,823.00
Gift
May 28, 2026 · 10:32 pm · @bob
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Bal $1,823.00
Welcome bonus
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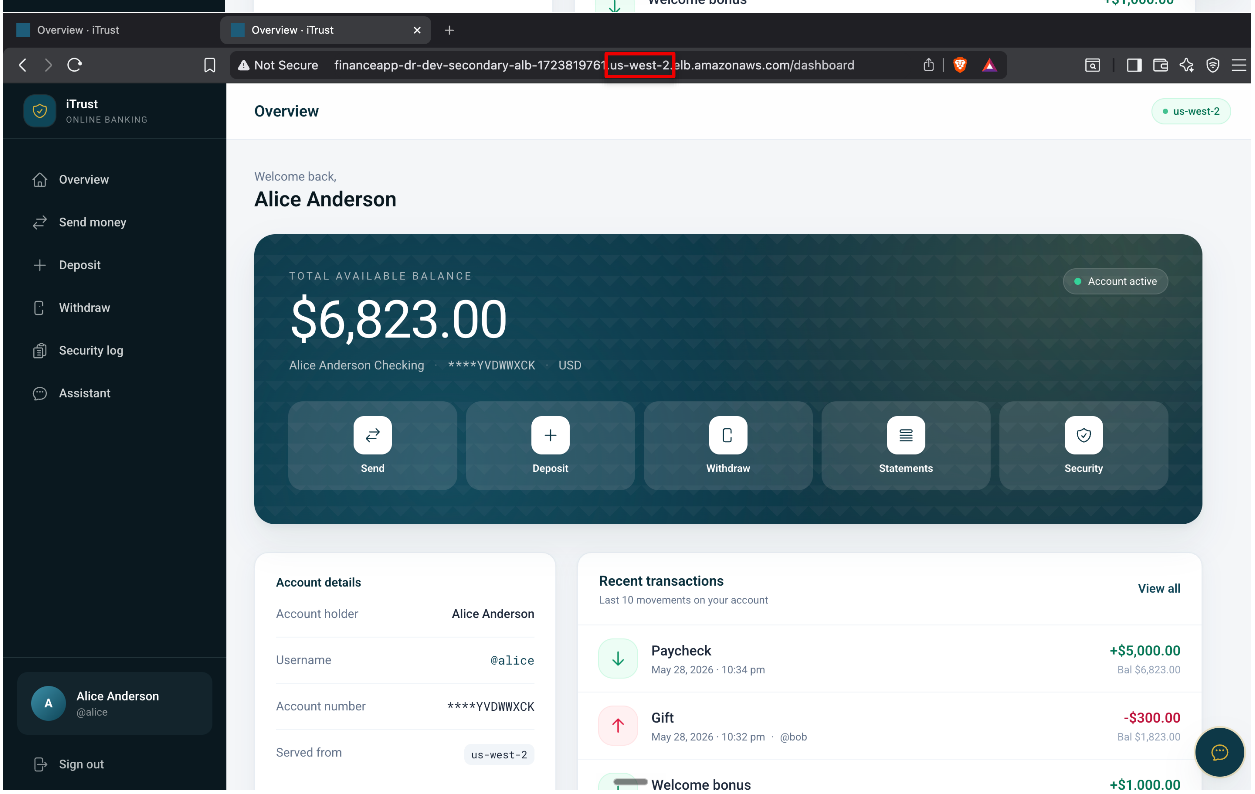
Alice Anderson
@alice
Sign out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Statements
Security
Account details
Account holder Alice Anderson
Username @alice
Account number ****YVDWWXCK
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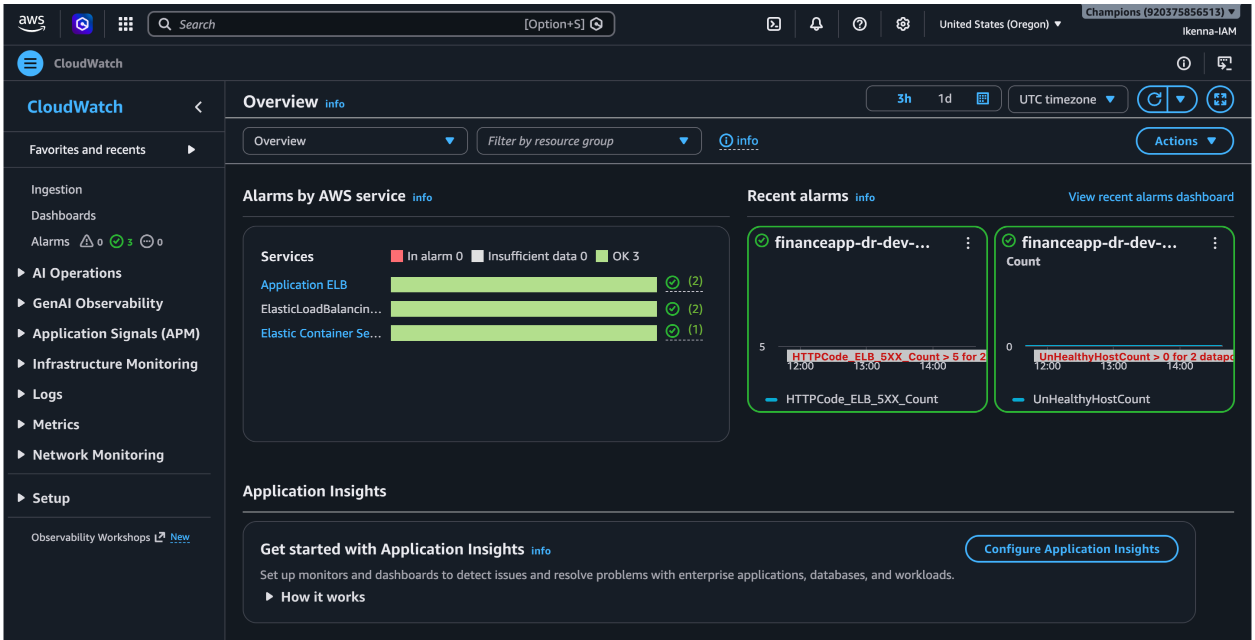

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Application Insights
Get started with Application Insights info
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Set up monitors and dashboards to detect issues and resolve problems with enterprise applications, databases, and workloads.
How it works
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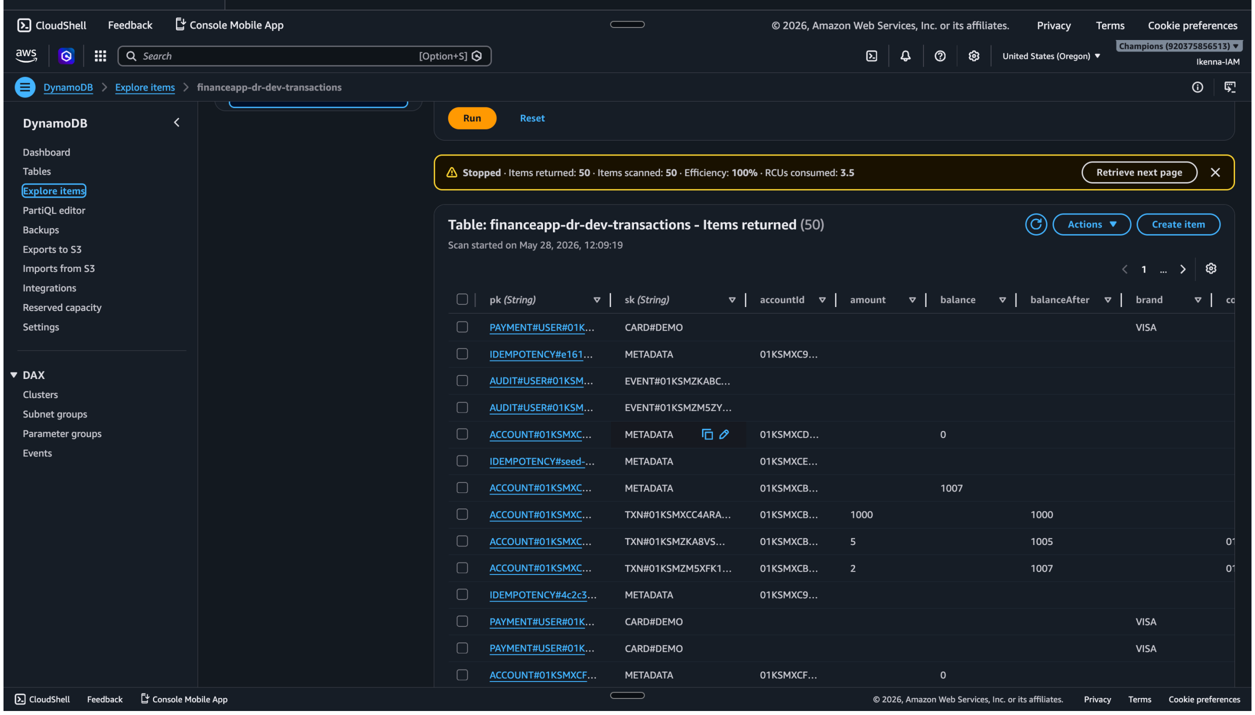

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DynamoDB
Dashboard
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Backups
Exports to S3
Imports from S3
Integrations
Reserved capacity
Settings
DAX
Clusters
Subnet groups
Parameter groups
Events
Run
Reset
Stopped · Items returned: 50 · Items scanned: 50 · Efficiency: 100% · RCUs consumed: 3.5
Retrieve next page
Table: financeapp-dr-dev-transactions - Items returned (50)
Scan started on May 28, 2026, 12:09:19
Actions
Create item
pk (String) | sk (String) | accountId | amount | balance | balanceAfter | brand
PAYMENT#USER#01K... | CARD#DEMO | | | | | VISA
IDEMPOTENCY#e161... | METADATA | 01KSMXC9...
AUDIT#USER#01KSM... | EVENT#01KSMZKABC...
AUDIT#USER#01KSM... | EVENT#01KSMZM5ZY...
ACCOUNT#01KSMXC... | METADATA | 01KSMXCD... | | 0
IDEMPOTENCY#seed-... | METADATA | 01KSMXCE...
ACCOUNT#01KSMXC... | METADATA | 01KSMXCB... | | 1007
ACCOUNT#01KSMXC... | TXN#01KSMXCC4ARA... | 01KSMXCB... | 1000 | | 1000
ACCOUNT#01KSMXC... | TXN#01KSMZKA8VS... | 01KSMXCB... | 5 | | 1005
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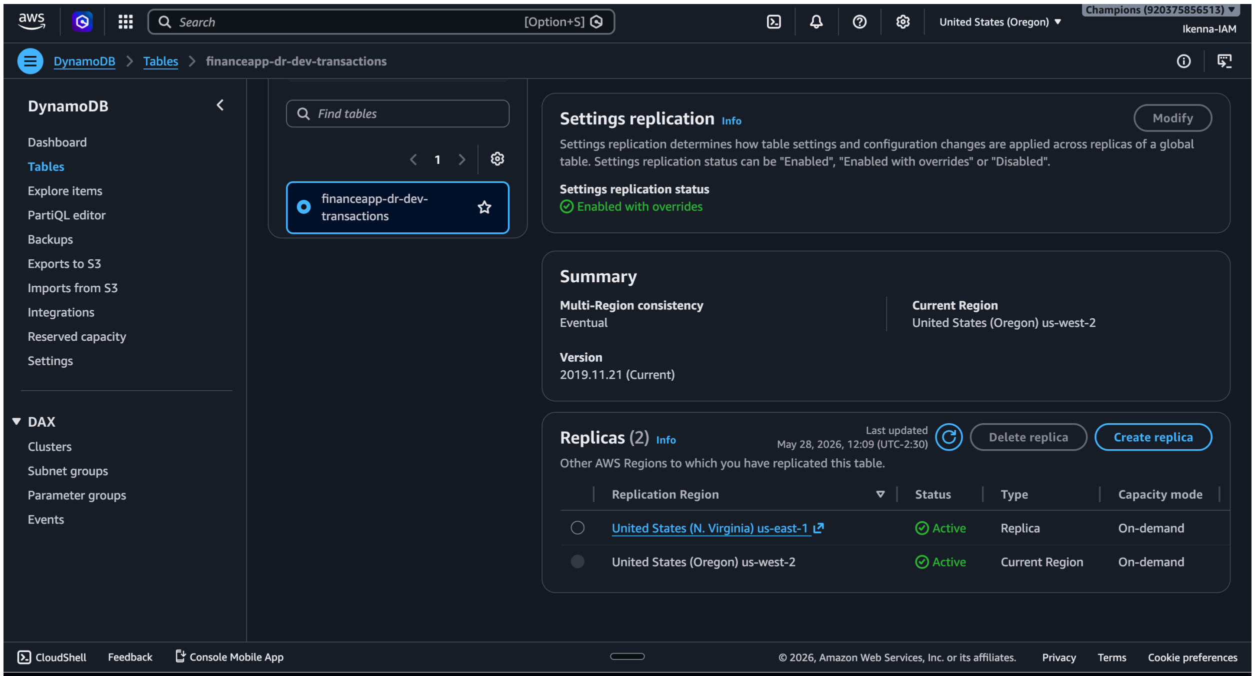

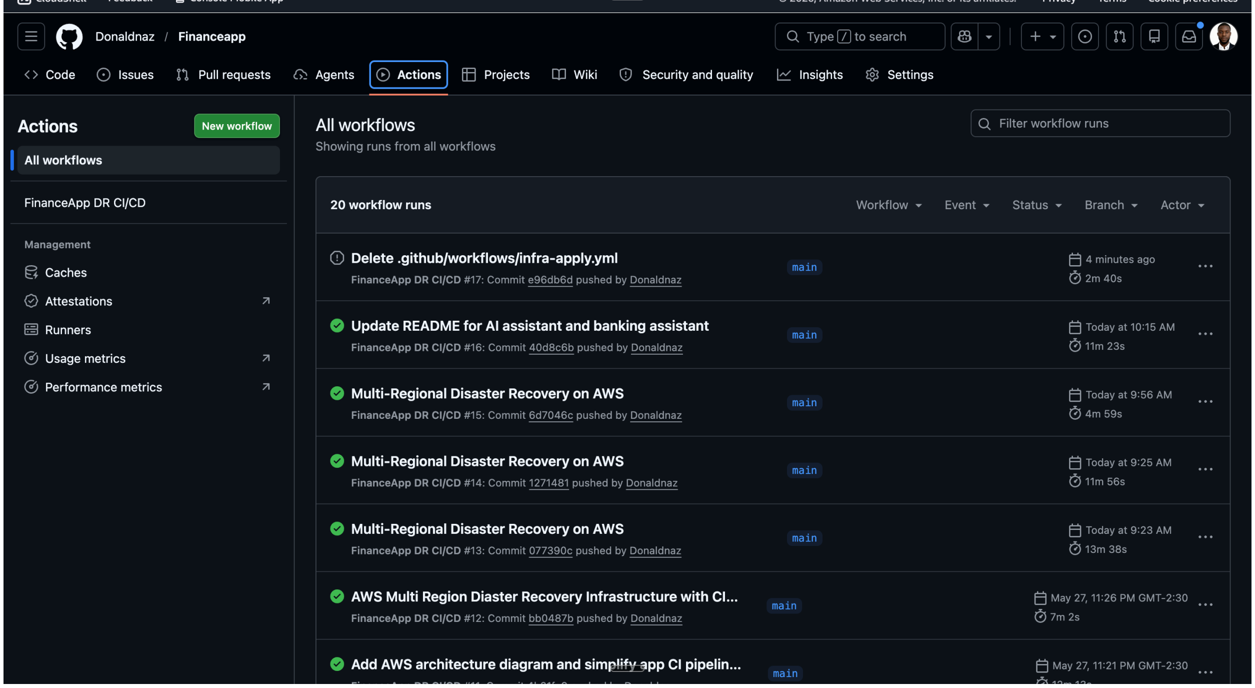

5. Business Impact

Transitioning from a traditional cloud posture to this multi-region declarative framework yields clear business and operational advantages:

- **Uninterrupted Business Continuity:** The application achieves true regional disaster isolation. Total failure of an entire AWS data-center hub triggers automated recovery procedures without service interruption for users.

- **Optimized Cost Efficiency:** By utilizing AWS Fargate serverless compute units and on-demand DynamoDB pricing models, the operational maintenance costs for running this multi-region infrastructure baseline are minimized to approximately $8.00 / day during idle testing.
- **Elimination of Manual Operations:** Infrastructure code guarantees environment parity. The exact topology can be reproduced instantly, eliminating manual execution variance or configuration drift during disaster scenarios.